Multi-Horizon, Multi-Entity Anonymous Time-Series Forecasting: Phase 1 Project Report — Advanced Machine Learning & Deep Learning, Sem-6

Based on: Hedge Fund — Time Series Forecasting (Kaggle competition, 2026)

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***Abstract*—**

This Phase 1 report outlines the foundational analysis and theoretical framework for the "Hedge Fund — Time Series Forecasting" competition. The project focuses on predicting synthetic financial instrument returns across multiple prediction horizons ($h \in \{1, 3, 10, 25\}$) using fully anonymised datasets containing an expanding universe of assets. We detail our extensive Exploratory Data Analysis (EDA), uncover severe violations of classical statistical assumptions (such as non-stationarity, fat-tailed distributions, and extreme multicollinearity), and propose a mathematically rigorous pipeline utilising robust loss landscapes, grouped median imputation, and horizon-normalised standardisations. Our findings mandate the use of tree-based ensembles and temporal sequence architectures over classical linear and univariate forecasting models. Crucially, every architectural and preprocessing decision is grounded in a critical review of the financial AI literature, whose most consistent finding is that evaluation rigour — not model complexity — determines genuine out-of-sample performance.

Index Terms—Time-Series Forecasting, Quantitative Finance, Machine Learning, Exploratory Data Analysis, Non-Stationarity, Walk-Forward Validation, Cross-Sectional Normalisation, Temporal Fusion Transformer.

I. INTRODUCTION

Quantitative finance frequently encounters extreme noise-to-signal ratios, structural distribution shifts, and temporal non-stationarity. In this project, heavily inspired by the Kaggle Hedge Fund Forecasting competition, we attempt to solve a multi-horizon, multi-entity cross-sectional forecasting problem. The dataset is heavily anonymised, comprised of arbitrary feature names, categorical entity identifiers (*code*, *sub_code*, *sub_category*), and temporal index sequences.

Three structural challenges distinguish this problem from standard time-series benchmarks studied in the literature. **First**, feature names (`feature_a` through `feature_ch`) carry no semantic content, ruling out the entire class of domain-engineered indicators — RSI, MACD, Bollinger Bands — that dominate the financial ML literature [3, 2]. **Second**, predictions must span four distinct forecast horizons across multiple (*code*, *sub-code*, *sub-category*) combinations, a hierarchical cross-sectional structure studied empirically by Roncalli [4] but rarely combined with strict causal evaluation constraints. **Third**, the competition enforces strict temporal causality (no look-ahead) and withholds 75% of test data for a private leaderboard — a design that directly exposes the evaluation gap Jiang [3] documents across 45 published studies: the majority of reported ML advantages evaporate under honest walk-forward validation.

This Phase 1 report defines the problem mathematically, explores the deep structural properties of the data, grounds every preprocessing and modelling choice in the reviewed literature, and designs a pipeline that resists catastrophic overfitting and feature leakage.

II. LITERATURE REVIEW

Seven papers are reviewed and organised into three thematic clusters. Rather than cataloguing methods, this review constructs a critical argument: the competition's evaluation design exposes the single most under-addressed flaw in the published literature — the systematic conflation of in-sample model complexity with genuine out-of-sample generalisation. Every modelling decision in Sections III–VI is derived from this body of work.

A. Cluster I: The Macro Landscape and Its Failure Modes

Cao (2022) [2] surveys financial AI across trading, risk management, and portfolio construction. Its contribution is its taxonomy of failure modes: (i) distributional non-stationarity; (ii) low signal-to-noise ratio; (iii) feature leakage in benchmark construction; (iv) absence of standardised evaluation protocols. Our EDA findings in Section IV directly confirm (i) and (ii) — the Frobenius-norm covariance drift and kurtosis of 290 are precisely the regime-instability and SNR failures Cao describes. The competition's sequential prediction requirement addresses (iii) by construction; addressing (iv) is the motivation for our GROUPTIMESERIESSPLIT validation protocol.

Jiang (2021) [3] conducts a meta-analysis of 45 stock market prediction studies. The key finding: LSTMs are reported as state-of-the-art in 62% of studies, but fewer than 23% use walk-forward validation and the LSTM advantage over gradient-boosted trees shrinks to within noise under honest evaluation. This is the **primary justification for choosing XGBoost/LightGBM as the Phase 1 baseline**: the literature's apparent LSTM superiority is a methodological artefact. Our EDA confirms this: maximum Pearson correlation below 0.05 places this dataset firmly in the low-SNR regime where tree en-

sembles with implicit feature subsampling are more robust than attention-based end-to-end models.

B. Cluster II: Cross-Sectional Prediction and Regime Robustness

Roncalli (2021) [4] studies predicting relative performance across a universe of entities at each time step — a cross-sectional task mirroring the (*code*, *sub-code*, *sub-category*) structure of this dataset precisely. Their central finding: **cross-sectional feature normalisation** — scaling features relative to the entity universe at each `ts_index` rather than using global statistics — substantially improves out-of-sample metrics. This directly motivates our horizon-based target normalisation (μ_h, σ_h) and per-timestamp feature standardisation. Roncalli also introduces a two-stage pipeline (regularised linear selection followed by modelling), which motivates our explicit noise-feature dropping and collinearity pruning before tree training.

Audrino et al. (2020) [5] compare AI-driven and human-managed hedge funds and find no statistically significant performance difference after controlling for strategy and leverage. The mechanism: AI models are exceptional in trained regimes and mediocre in others. Our EDA discovery of 35 entirely novel *sub_codes* in the unseen test set is a direct empirical manifestation of this regime-shift risk. The consequence: grouping by high-level invariants (*code* $\times$ *sub_category*) to prevent catastrophic variance on unseen sub-codes follows directly from Audrino's analysis.

A direct tension between Roncalli and Audrino is worth making explicit: cross-sectional ML works (Roncalli) but fails across regimes (Audrino). Our synthesis: cross-sectional normalisation is a robust preprocessing step that survives regime change, while model weights are regime-specific and must be validated with expanding windows, not fixed splits.

C. Cluster III: Deep Learning, LLMs, and RL — Scope and Limits

Zhang et al. (2023) [6] trace the progression from classical factor models through Transformers to LLMs as financial signal extractors. Their key finding: the *Temporal Fusion Transformer* (TFT) of Lim et al. [7] outperforms LSTMs on multi-horizon benchmarks by learning horizon-specific attention patterns. This is directly relevant to our $h \in \{1, 3, 10, 25\}$ structure and justifies the Phase 2 TFT roadmap. Critically, every LLM result in the paper relies on news text or earnings filings — all forbidden here. Zhang et al. is cited for its sequence-model architecture survey only.

Fischer (2018) [8] surveys RL in finance. The critical warning: every successful RL result is achieved in simulation with fixed dynamics. Policy gradient methods also require substantially more environment interactions than supervised methods require data samples. Given our low-SNR regime and data constraints, RL cannot serve as a primary forecasting method. It is retained as a future extension for portfolio allocation only.

Jiang (2021) + López de Prado (2018) [3, 1] together motivate two-stage feature selection under low-SNR conditions. López de Prado's combinatorial purged cross-validation formalises the look-ahead bias problem Jiang identifies empirically, providing the theoretical grounding for GroupTimeSeriesSplit.

D. Literature-to-Pipeline Derivation

Table 1 makes the literature-to-implementation chain explicit.

Table 1. Literature-to-Project Design Decision Derivation

Paper	Key Finding	Phase 1 Decision	Section
Cao [2]	Eval. gaps & SNR	GroupTimeSeriesS	V
Jiang [3]	LSTM = artefact	XGB/LGB baseline	VI
Roncalli [4]	X-sect. norm	Horizon μ_h, σ_h norm	VI
Audrino [5]	Regime collapse	Group by code×sub_cat	V
Zhang [6]	TFT multi-horizon	Phase 2 TFT roadmap	VII
Fischer [8]	RL sample-ineff.	RL excluded (Ph.2 ext.)	VII
L. de Prado [1]	Purged CV	Expanding window splits	V

III. PROBLEM FORMULATION

The objective is to establish a robust predictive framework. Mathematically, the task is a sequence-to-scalar cross-sectional prediction problem. Given a multidimensional time-series covariate matrix $X_t \in \mathbb{R}^{N_t \times F}$ at time t, where N_t is the expanding universe of entities and $F = 86$ anonymised features, our goal is to predict the target variable Y_{t+h} over multiple horizons $h \in \{1, 3, 10, 25\}$.

Unlike univariate forecasting (i.e., standard ARIMA), this is a cross-sectional panel data problem governed by overlapping return windows. The target variance scales roughly proportional to $\sqrt{h}$ — behaving akin to a geometric Brownian motion. Standard MSE is insufficient, as it fails to account for horizon-dependent volatility, heavily biasing the model towards the $t+25$ outputs. The competition metric is a weighted RMSE skill score:

$$S = \sqrt{1 - \text{clip}\left(\frac{\sum_{i \in I} w_i (y_i - \hat{y}_i)^2}{\sum_{i \in I} w_i y_i^2},\ 0,\ 1 \right)} \tag{1}$$

where w_i are per-row competition weights (used in evaluation only, not as training features), and I is the evaluation index set. A model predicting all zeros scores $S = 0$; a perfect model scores $S = 1$. The denominator $\sum w_i y_i^2$ means the metric is scale-sensitive and punishes horizon-specific bias — directly motivating the horizon normalisation in Section VI.

IV. EXPLORATORY DATA ANALYSIS

Our EDA, structured across three iterative computational notebooks, revealed fundamental dynamics regarding the target signal and feature structure, confirming the failure modes documented by Cao [2] and Jiang [3] at the dataset level.

A. Target Variable Properties

The target variable exhibits extreme leptokurtosis, with a kurtosis of approximately 290. Values are heavily concentrated around zero but feature massive, unpredictable tails. Shapiro-Wilk and Anderson-Darling tests conclusively rejected normality. An expanding universe size means earlier periods contain sparse entity counts, directly impacting cross-validation sampling mechanisms. This fat-tail structure is precisely the low-SNR, high-kurtosis regime that Cao [2] identifies as the primary driver of out-of-sample model degradation — standard L_2 loss under kurtosis of 290 will catastrophically over-weight rare tail events, producing a model that fits the noise rather than the signal.

B. Feature Space and Signal-to-Noise Ratio

The dataset has an exceptionally low signal-to-noise ratio: the highest absolute Pearson correlation with the target is below 0.05. We discovered heavy nonlinear and monotonic dependencies using Spearman rank and Mutual Information arrays. Hierarchical clustering highlighted massive redundancy, isolating nearly perfect duplicates (e.g., $r > 0.95$ for variables such as *bm/bo* and *bz/cd*). Completely uninformative features acting as injected random noise (e.g., `feature_b` to `feature_g`) were also identified.

This directly validates Jiang's [3] finding that end-to-end deep feature learning is unreliable in low-SNR financial regimes, and Roncalli's prescription [4] of an explicit selection stage before deep model training. With maximum Pearson correlation below 0.05, attention mechanisms on 86 anonymous features will amplify noise, not suppress it.

C. Temporal Distribution Shifts

We detected highly structural concept drift based on rolling covariance matrices calculated via the Frobenius norm. Kolmogorov-Smirnov (KS) tests on specific features exposed deep distribution shifts between train and test boundaries. Additionally, 35 entirely novel *sub_codes* appear in the unseen test dataset, creating a zero-shot generalisation requirement.

This is the empirical manifestation of Audrino et al.'s [5] regime-instability finding: models trained on the observed *sub_code* distribution encounter a genuinely different cross-sectional universe at test time. Grouping by *code* $\times$ *sub_category* to ensure generalisation to unseen sub-codes follows directly from Audrino's analysis of where AI model variance collapses in novel regimes.

V. THEORETICAL RIGOUR & STATISTICAL ANALYSIS

Applying basic models to this dataset blindly violates multiple statistical theorems. This section links our empirical findings to strict theoretical constraints and to the corresponding literature-grounded design decisions.

A. Violations of I.I.D. Assumptions

Because the covariance matrix fluctuates wildly and the entity count N_t monotonically increases, the underlying data-generating process is not identically distributed. Standard K-fold cross-validation causes temporal feature leakage — precisely the failure mode (iii) documented by Cao [2] and formalised theoretically by López de Prado's combinatorial purged cross-validation [1]. We mandate an expanding moving-window approach (GROUPTIMESERIESSPLIT) explicitly aligned with the *ts_index*. This directly implements the walk-forward evaluation that Jiang [3] identifies as the critical missing element in 77% of published financial ML studies.

B. Fat Tails and Robust Optimisation

The massive tail events distort standard L_2 objectives. Incorporating Huber Loss, Mean Absolute Error (L_1), or explicit 1st/99th percentile target clipping is mandatory. Furthermore, the instance weights span 13 orders of magnitude; optimising directly with these weights triggers exploding gradients, demanding weight decay constraints or normalised batch routing during stochastic gradient descent. Since the skill score denominator in Eq. (1) is $\sum w_i y_i^2$, outlier targets under high weights dominate the gradient landscape. Robust loss functions are therefore a theoretical necessity given confirmed kurtosis of 290, not a hyperparameter choice.

C. Bias-Variance Tradeoff on Hierarchical Data

A primary challenge is zero-shot generalisation. Our distribution analysis demonstrated 35 entirely novel *sub_codes* in the unseen dataset. Overfitting high-capacity neural networks to arbitrary test sub-codes generates catastrophic variance. The hypothesis space must be generalised via grouping by high-level invariants (*code* and *sub_category*). This is the pipeline-level implementation of Audrino et al.'s [5] finding that ML model variance is highest in novel regimes: we accept higher bias in exchange for controlled variance on the 35 unseen sub-codes.

VI. PROPOSED ARCHITECTURE & PREPROCESSING PIPELINE

To respect our theoretical observations and the literature-grounded design decisions in Table 1, the Phase 1 implementation (defined in `pipeline/`) enforces the following:

1. **Dimensionality Compression.** Noise features are explicitly dropped early in the preprocessing configuration (`preprocess_config.json`). Collinear features are pruned based on cluster distances. This implements the first stage of Roncalli's two-stage pipeline [4]: regularised selection before modelling, grounded in our EDA finding of near-perfect duplicates ($r > 0.95$) and injected-noise features with zero mutual information.

2. **Missing Entity Logic.** Due to high sparsity matrices for certain entity sub-codes, we employ grouped median

imputation (*code* × *sub_category*) while inserting binary `_is_missing` boolean flags, forcing models to utilise structural emptiness as a geometric predictor. Imputing within *sub_category* rather than *sub_code* is grounded in Audrino's [5] regime-instability finding: the *sub_code*-level distribution does not generalise to the 35 unseen sub-codes at test time.

3. **Horizon-Based Target Normalisation.** The target is standardised across each independent forecasting horizon (μ_h, σ_h) to preserve equal gradient weighting across $h = 1$ and $h = 25$. This directly implements Roncalli's cross-sectional normalisation principle [4] adapted to the horizon dimension: since target variance scales with $\sqrt{h}$, global normalisation systematically biases training toward long-horizon outputs. Per-horizon standardisation removes this bias, aligning with the competition metric in Eq. (1).

4. **Model Archetypes.** The initial test bed incorporates gradient boosted frameworks (XGBoost, LightGBM) that handle multicollinearity automatically via subspace sampling and possess invariance to monotonic rank shifts. This choice is grounded directly in Jiang [3]: under walk-forward evaluation — enforced by construction through the competition's sequential prediction requirement — tree ensembles are competitive with LSTMs, faster to tune, and yield interpretable SHAP rankings that allow us to audit the anonymous feature space.

5. **Validation Protocol.** GROUPTIMESERIESSPLIT with expanding windows ensures no temporal leakage. Following Cao's [2] framework, the 75% private leaderboard means the public 25% is treated as a diagnostic only, not as a model-selection criterion.

VII. CONCLUSION & PHASE 2 ROADMAP

Phase 1 successfully scopes the mathematical hostility of the anonymised hedge fund dataset. We have mapped target fat-tails (kurtosis ≈ 290), documented feature drift via Frobenius-norm covariance analysis and KS tests, identified 35 zero-shot sub-codes in the test set, engineered a logically sound validation protocol resisting future leakage (GROUPTIMESERIESSPLIT), and constructed a robust data ingestion pipeline with grouped median imputation, horizon normalisation, and noise-feature elimination. Each decision is derived from a specific finding in the reviewed literature rather than adopted as a default heuristic.

The central argument, grounded in the literature review, is that the competition's evaluation design directly tests the property the literature most consistently fails to demonstrate: genuine out-of-sample generalisation under regime change. Phase 1 addresses this by prioritising evaluation discipline over model complexity.

Phase 2 will:

1. Test LightGBM with Huber/MAE loss and Optuna hyperparameter tuning across specialised horizon-subcategory models.
2. Implement SHAP-based feature pruning from the ensemble baseline before deep model training — the second stage of Roncalli's pipeline [4].
3. Explore the Temporal Fusion Transformer (TFT) [7, 6] for auto-correlative momentum detection across the four horizons, leveraging its horizon-specific attention mechanism suited to $h \in \{1, 3, 10, 25\}$.
4. Implement regime-stratified performance reporting (stable vs. volatile target-distribution periods) per Audrino et al. [5] and a rolling ensemble combining models trained on different temporal windows.
5. Evaluate an RL-based portfolio allocation layer over point forecasts as a future extension, acknowledging Fischer's [8] sample efficiency constraints as the primary risk.

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